



Guidelines

Naloxone

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Hospital-Wide	Medical, Nursing, Pharmacy

Introduction

This guideline provides guidance to all Sir Charles Gairdner Hospital (SCGH) staff involved in the prescription, administration, supply and management of patients on naloxone. This document supports and complies with related legislation, policy and other related documents [here](#).

1.0 General Information

Naloxone is a competitive opioid antagonist at mu, kappa and delta receptors. It is used to partially or completely reverse the adverse effects of natural and synthetic opioids such as sedation and respiratory depression. It has little effect when given orally (high first pass effect) or sublingually. Opioid side effects such as drowsiness, respiratory depression or pruritus may be reversed with low doses of naloxone whilst maintaining analgesia. This is because analgesia occurs with lower plasma concentrations than those that are usually required to cause these side effects. However, analgesia will also be affected if the dose of naloxone administered is increased.

2.0 Presentation

Naloxone 400microg/mL Ampoule

3.0 Indication

- Complete or partial reversal of opioid-induced respiratory or CNS depression
- Suspected acute opioid overdose
- Opioid-induced pruritus, especially if due to epidural or intrathecal opioids.
 - Naloxone reduces the incidence of opioid induced pruritus and nausea but not vomiting.

Out of Scope: Naloxone infusion for cholestatic pruritis is not addressed in this guideline

4.0 Contraindications

Hypersensitivity to naloxone

5.0 Precautions

Physical opioid dependence

- Acute opiate withdrawal may be precipitated when a supra-therapeutic dose has been administered to such patients
- Close observation for signs and symptoms of withdrawal is warranted during titration

Long-acting opiates

- The duration of action of naloxone may be shorter than that of some opioids
- Observation for recurrence of unwanted opiate effects should occur for a minimum of 2 hours after the last dose of naloxone

Pain precipitation

- Naloxone may precipitate pain in patients with chronic or acute pain issues who have been, or currently are stabilised on opioid analgesia

Non-opiate induced respiratory/CNS depression

- Naloxone can be effective in respiratory or CNS depression from other causes due to reversal of endogenous opiates
- Care should be taken that this clinical improvement should not be interpreted as drug induced opiate toxicity

Reversal of buprenorphine-induced respiratory depression may be incomplete and may require very high doses of naloxone

6.0 Dose and Administration

Therapeutic doses of naloxone range from 40 micrograms to 400 micrograms, though larger doses may be required where the overdose is due to partial antagonists (up to 12mg may be required to reverse buprenorphine overdose). Naloxone is best administered via the IV route due to its rapid onset of action and for ease of dose titration, though may be given via IM, subcutaneous or intranasal routes if IV access is not established.

Titration of naloxone is vital, sudden or complete reversal of opioid toxicity may cause agitation and acute opiate withdrawal in opioid-dependent patients. To avoid acute withdrawal, use small, titrated doses given every 2 to 3 minutes.

Titration of naloxone dose ensures that analgesia effects are not reversed especially for patients known to Pain Services at Sir Charles Gairdner Hospital

6.1 Bolus dose for Respiratory/CNS Depression associated with opioids

- Dilute 400 micrograms Naloxone to 10mL with Sodium Chloride 0.9% to make a 40 micrograms/mL solution.
- Administer a bolus dose of 40 micrograms (1mL) IV; repeat every 2 minutes until symptoms/signs are resolved.

If continuous naloxone therapy is required for reversal of respiratory/CNS depression for suspected acute opioid overdose, commence an intravenous naloxone infusion

- Patients requiring IV naloxone infusion require transfer to a designated high dependency unit/critical care area.
- Dilute 2mg (5 ampoules) Naloxone in 100mL Sodium Chloride 0.9% to make a concentration of 20microgram/mL
- Commence the hourly infusion rate at two-thirds of the initial total effective bolus dose required in the first hour to elicit a response.
 - For example: If a total of 120 micrograms was required via bolus injections to elicit a clinical response, start the infusion at two-thirds of 120 micrograms/hour, i.e. at 80micrograms/hour (equivalent to 4mL/hour if infusion is made as described above).
- The subsequent rate should be titrated to the patient's clinical response

- e.g. respiratory rate, oxygen saturation, end-tidal partial pressure of carbon dioxide, sedation score or signs of withdrawal
- Anticipate ongoing naloxone administration following overdose of methadone or controlled-release formulations.
- Gradually reduce the dose based on clinical symptoms when stopping the infusion, to accommodate opioids which have a longer half-life than naloxone (see Pharmacokinetics below).

6.2 Continuous infusion for reversal of respiratory depression/CNS depression whilst retaining analgesia effects

- Contact the Acute Pain Service in hours via pager 4120 or for afterhours contact the Anaesthesia Registrar on call via pager 4823 for advice.
- Commence the hourly infusion at 0.25microg/kg/hr (i.e. 20micrograms/hour or 1mL/hour in an 80kg person)
- The infusion should be titrated to affect. To avoid side effects, a maximum infusion rate of 1microg/kg/hour is recommended.
- Higher doses of naloxone are required for reversal of buprenorphine induced respiratory depression. (In this case it is recommended to follow the above dosing advice for acute opioid overdose)
- Review opioid analgesic order and optimise non-opioid analgesia
- Consider ceasing or withholding all other infusions/medications that may be contributing to sedation and or respiratory depression (e.g. benzodiazepines)
- Consideration should be given to cessation of the naloxone infusion after 24 hours or after 4 terminal half-lives of the opioid.

6.3 For pruritus treatment following intrathecal/epidural morphine

Note: For pruritus treatment naloxone is administered via the subcutaneous route

- Dilute 400 micrograms Naloxone to 10mL with Sodium Chloride 0.9% to make a 40 micrograms/mL solution.
- Administer 100 micrograms (2.5mL) by subcutaneous injection every 2 hours as required. Maximum of 1200microg per day unless under guidance of Acute Pain Service.
- Any unused naloxone must be discarded.

6.4 Pharmacokinetics

- The onset of action of naloxone after intravenous administration is about 1 minute.
- Naloxone is also well absorbed after IM, subcutaneous and intranasal administration.
- The duration of action of naloxone is relatively short (15 to 30 minutes), with a half-life in adults of approximately 1 hour.
- The duration of effect also depends on the dose and rate of elimination of the agonist, hence the possible requirement of infusions given most opioids have longer durations of action compared to naloxone.

7.0 Monitoring

7.1 Monitoring Intravenous

- Document respiratory rate, oxygen saturation and consciousness 5-minutely for the first hour, then half-hourly for the second hour. All observations should be recorded

on the Adult Observation and Record Chart. Critical care area observations may occur 10-15 minutely in the first hour

- If infusion continues after two hours, then continue with hourly respiratory rate, sedation scores and four-hourly blood pressure and pulse until two hours after the infusion is ceased
- Check infusion and document pain score on Observation and Response Chart hourly for the first four hours, then four-hourly thereafter
- Monitor patient for signs of opioid withdrawal and/or pain (if there is a history of acute/chronic pain)
- Complications associated with opiate withdrawal should be managed accordingly
- All patients given naloxone should be observed for re-sedation for at least 2 hours after the last dose

7.2 Monitoring Subcutaneous

- Reassess patient's pruritus (verbally) every 4 hours whilst awake. Escalate for medical review if not improving

7.3 Nursing Management

- Naloxone has a short duration of effect, if the IV cannula tissues contact the treating team for urgent re-siting.
- Referral to the Drug and Alcohol nursing team may be warranted if illicit substance abuse or prescribed opioid intentional misuse is suspected.

8.0 Adverse Effects

- Common: erythema at injection site, dizziness, headache
- Opioid withdrawal: In opioid-tolerant patients, dose-dependent production of a withdrawal syndrome occurs. This can include severe agitation and aggression and should be avoided. If withdrawal syndrome is inadvertently produced, immediately cease further administration of naloxone. If reassurance fails, physical and chemical control of the patient may be necessary. Avoid long-acting chemical sedation as the duration of naloxone-induced withdrawal is short (usually <90 minutes).
- Rapid, complete reversal by naloxone can potentially cause serious adverse effects such as cardiac arrest, hypertensive crisis, hypotension, ventricular arrhythmias, coma, encephalopathy, seizures, and pulmonary oedema

9.0 Compatible Fluids

- Glucose 5%, sodium chloride 0.9%
- Refer to the [Australian Injectable Drugs Handbook](#) for further information

10.0 Pregnancy and Breastfeeding

- If acute opioid toxicity is evident in a pregnant woman, treatment with naloxone should not be withheld. Monitoring of infant heart rate is recommended.
- Naloxone is considered safe to use in breastfeeding; though it may pass into breast milk, it has low oral bioavailability. It is recommended that breastfed infants of opioid-dependent women are monitored for signs of withdrawal
- Contact the Obstetric Medicines Information Service specialist advice on 6458 2723

Related Documents

Legislation:

- [Medicines and Poisons Act 2014](#)
- [Medicines and Poisons Regulations 2016](#)
- [Health Services Act 2016](#)

Authorising Policy and Standards:

- [SCGOPHCG Medicines Management Policy and Procedure](#)
- [Mandatory Policy MP0077/18 Statewide Medicines Formulary Policy](#)
- [SCGOPHCG High Risk Medication Policy](#)
- [OD 0647/16 National Standard for User-Applied Labelling of Injectable Medicines, Fluids and Lines 2016](#)
- NSQHS Standards 1, 4, 8

Standards, procedures, guidelines:

- [Intravenous Therapy. SCGH Nursing Practice Guideline 4](#)
- [Nursing Practice Guideline No. 51 – Medication Management](#)

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For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

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